Fourier

Law:-

$$F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{T}\right) + b_n \sin\left(\frac{n\pi x}{T}\right)\right)$$

Odd: (F(x) odd , Fourier sine series.)

$F(x) = x$

$F(-x) = -x$

$F(x) = -F(-x)$ (odd)

$a_0 = a_n = 0$

$b_n = \frac{2}{T}\int_0^T F(x)\sin\left(\frac{n\pi x}{T}\right)dx$

$F(x) = \sum_{n=1}^{\infty}\left(b_n \sin\left(\frac{n\pi x}{T}\right)\right)$

Even: (F(x) even, Fourier cosine series.)

$F(x) = x^2$

$F(-x) = (-x)^2 = x^2$

$F(x) = F(-x)$ (even)

$b_n = 0$

$a_0 = \frac{2}{T}\int_0^T F(x)dx$

$a_n = \frac{2}{T}\int_0^T F(x)\cos\left(\frac{n\pi x}{T}\right)dx$

$F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{T}\right)$

$* \sin(n\pi) = 0$, $* \cos(n\pi) = (-1)^n$

$F(x): -T \le x \le T \quad \rightarrow \quad T - (-T) = 2T$

$Ex: F(x): -\pi \le x \le \pi$

$(T - -T) = (\pi - -\pi) \quad \rightarrow \quad 2T = 2\pi \quad \rightarrow \quad T = \pi$

Ex: $F(x) = x$ $\quad -\pi \le x \le \pi$

$F(x) = -F(-x)$ $\quad odd$, $\quad 2T = 2\pi \quad \rightarrow \quad T = \pi$

$$a_0 = a_n = 0 \quad , \quad F(x) = \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$b_n = \frac{2}{T}\int_0^T F(x)\sin\left(\frac{n\pi x}{T}\right)dx$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} x\ \sin\left(\frac{n\cancel{\pi}x}{\cancel{\pi}}\right)dx = \frac{2}{\pi}\int_0^{\pi} x\ \sin(nx)\,dx$$

$$b_n = \frac{2}{\pi}\left[\frac{-x\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right]\pi, 0$$

$$b_n = \frac{2}{\pi}\left[\frac{-\pi\cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2}\right] - 0$$

$$b_n = \frac{2}{\pi}\left[\frac{-\pi(-1)^n}{n} + \frac{0}{n^2}\right] = \frac{2}{\pi}\left(\frac{-\pi(-1)^n}{n}\right)$$

x	$+$	$\sin(nx)$
1	$-$	$-\frac{\cos(nx)}{n}$
0		$-\frac{\sin(nx)}{n^2}$

Ex: $F(x) = x$ $\quad$ *Fourier cosine* $\quad 0 \le x \le \pi$

$cosine \rightarrow even$, $\quad T = \pi$

$$b_n = 0\ , a_0 = \frac{2}{\pi}\int_0^{\pi} x\,dx \ , \ a_n = \frac{2}{\pi}\int_0^{\pi} x\cos\left(\frac{n\cancel{\pi}x}{\cancel{\pi}}\right)dx = \frac{2}{\pi}\int_0^{\pi} x\cos(nx)\,dx$$

$$F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos(nx)$$

$$a_0 = \frac{2}{\pi}\left(\frac{x^2}{2}\right)\pi, 0 = \frac{2}{\pi}\left(\frac{\pi^2}{2}\right)$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x\cos(nx)\,dx = \frac{2}{\pi}\left[\frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right]\pi, 0$$

$$a_n = \frac{2}{\pi}\left[\frac{\pi\sin(n\pi)}{n} + \frac{\cos(n\pi)}{n^2}\right] = \frac{2}{\pi}\left((0 + \frac{(-1)^n}{n^2}) - \frac{1}{n^2}\right) = \frac{2}{\pi}\left(\frac{(-1)^n}{n^2} - \frac{1}{n^2}\right)$$